

1. NAME OF THE MEDICINAL PRODUCT

MAXIDEX ophthalmic suspension

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

1 ml of suspension contains 1 mg dexamethasone.

Excipient with known effect in suspension: benzalkonium chloride 0.1 mg/ml.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Eye drops, suspension.

Opaque, white to pale yellow suspension, no agglomerates.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

MAXIDEX contains dexamethasone, a synthetic corticosteroid.

MAXIDEX is indicated in the management of conditions generally responsive to corticosteroids such as:

- Certain inflammatory eye conditions of the anterior segment: acute and chronic anterior uveitis, iridocyclitis, iritis and cyclitis, herpes zoster ophthalmicus.
- Certain external diseases such as phlyctenular kerato-conjunctivitis, non-purulent conjunctivitis, including vernal, allergic, catarrhal. It is very effective where allergy is a main factor.
- Recurrent marginal ulceration of toxic or allergic etiology.
- Thermal and chemical burns.
- Post-operatively to reduce inflammatory reactions.

4.2 Posology and method of administration

Posology

- Topical application (1 or 2 drops in the conjunctival sac).
- SEVERE OR ACUTE INFLAMMATION: Every 30 to 60 minutes as initial therapy, reducing the dosage when favourable response is observed to every 2 to 4 hours. Further reduction may be made to 1 drop 3 or 4 times daily if sufficient to control inflammation. If favourable response is not obtained in 3 to 4 days, additional systemic or conjunctival therapy may be indicated.
- CHRONIC INFLAMMATION: Every 3 to 6 hours, or as frequently as necessary. Being tapered to discontinuation as inflammation subsides.
- ALLERGIES OR MINOR INFLAMMATION: Every 3 to 4 hours until the desired response is obtained. Being tapered to discontinuation as inflammation subsides.

Method of administration

For ocular use.

Shake the bottle well before use.

After cap is removed, if tamper evident snap collar is loose, remove before using product.

Do not let the tip of the dropper touch the eye.

Nasolacrimal occlusion or gently closing the eyelid after administration is recommended. This may reduce the systemic absorption of medicinal products administered via the ocular route and result in a decrease in systemic adverse reactions.

If more than 1 topical ophthalmic medicinal product is being used, the medicines must be administered at least 5 minutes apart. Eye ointments should be administered last.

Special populations

Pediatric population

The safety and efficacy of MAXIDEX ophthalmic suspension in children have not been established.

Geriatric population

No overall differences in safety or effectiveness have been observed between elderly and younger patients.

Hepatic and renal impairment

The safety and efficacy of MAXIDEX ophthalmic suspension in patients with hepatic or renal impairment have not been established.

4.3 Contraindications

- Hypersensitivity to the active substance or to any of the excipients.
- Acute, untreated bacterial infections.
- Herpes simplex keratitis.
- Vaccinia, varicella, and other viral infections of cornea or conjunctiva.
- Fungal diseases of ocular structures or untreated parasitic eye infections.
- Mycobacterial ocular infections.

4.4 Special warnings and precautions for use

- Nasolacrimal occlusion or gently closing the eyelid after administration is recommended. This may reduce the systemic absorption of medicinal products administered via the ocular route and result in a decrease in systemic adverse reactions.
- Prolonged use of topical ophthalmic corticosteroids may result in ocular hypertension and/or glaucoma, with damage to the optic nerve, reduced visual acuity and visual field defects, and posterior subcapsular cataract formation. In patients receiving prolonged ophthalmic corticosteroid therapy, intraocular pressure should be checked routinely and frequently. This is especially important in pediatric patients, as the risk of corticosteroid-induced ocular hypertension may be greater in children and may occur earlier than in adults. MAXIDEX is not approved for use in pediatric patients. The risk of corticosteroid-induced raised intraocular pressure and / or cataract formation is increased in predisposed patients (e.g. diabetes).
- Cushing's syndrome and/or adrenal suppression associated with systemic absorption of ophthalmic dexamethasone may occur after intensive or long-term continuous therapy in predisposed patients, including children and patients treated with CYP3A4 inhibitors (including ritonavir and cobicistat). (See Section 4.5). In these cases, treatment should not be discontinued abruptly, but progressively tapered.
- Corticosteroids may reduce resistance to and aid in the establishment of bacterial, viral, fungal or parasitic infections and mask the clinical signs of infection.
- Fungal infection should be suspected in patients with persistent corneal ulceration. Corticosteroids therapy should be discontinued if fungal infection occurs.

- Topical ophthalmic corticosteroids may slow corneal wound healing. Topical NSAIDs are also known to slow or delay healing. Concomitant use of topical NSAIDs and topical steroids may increase the potential for healing problems (see section 4.5).
- In those diseases causing thinning of the cornea or sclera, perforations have been known to occur with the use of topical corticosteroids. The wearing of contact lenses is discouraged during treatment of an ocular inflammation. MAXIDEX ophthalmic suspension contains benzalkonium chloride which may cause eye irritation and is known to discolour soft contact lenses. Avoid contact with soft contact lenses. However, if the healthcare provider considers contact lenses use appropriate, patients must be instructed to remove contact lenses prior to application of MAXIDEX ophthalmic suspension and wait at least 15 minutes before reinsertion.

4.5 Interaction with other medicinal products and other forms of interaction

- Concomitant use of topical steroids and topical NSAIDs may increase the potential for corneal healing problems. CYP3A4 inhibitors including ritonavir and cobicistat may increase systemic exposure resulting in increased risk of adrenal suppression/Cushing's syndrome. (See Section 4.4). The combination should be avoided unless the benefit outweighs the increased risk of systemic corticosteroid side-effects, in which case patients should be monitored for systemic corticosteroid effects.

4.6 Pregnancy and lactation

Fertility

Studies have not been performed to evaluate the effect of topical ocular administration of dexamethasone on fertility. There is limited clinical data to evaluate the effect of dexamethasone on male or female fertility. Dexamethasone was free of adverse effects on fertility in a chorionic gonadotropin primed rat model.

Pregnancy

There are no adequate or well-controlled studies evaluating MAXIDEX ophthalmic suspension in pregnant women. Prolonged or repeated corticoid use during pregnancy has been associated with an increased risk of intra-uterine growth retardation. Infants born of mothers who have received substantial doses of corticosteroids during pregnancy should be observed carefully for signs of hypoadrenalism (See Section 4.4). Studies in animals have shown reproductive toxicity after systemic administration. The ocular administration of 0.1% dexamethasone also resulted in fetal anomalies in rabbits (see section 5.3).

MAXIDEX ophthalmic suspension is not recommended during pregnancy.

Lactation

It is unknown whether MAXIDEX ophthalmic suspension is excreted in human milk. No data is available on the passage of dexamethasone into human breast milk. It is not likely that the amount of dexamethasone would be detectable in human milk or be capable of producing clinical effects in the infant following maternal use of the product.

A risk to the breastfed child cannot be excluded. A decision must be made whether to discontinue breast-feeding or to discontinue or abstain from MAXIDEX therapy taking into account the benefit of breast feeding for the child and the benefit of therapy for the woman.

4.7 Effects on ability to drive and use machines

Temporary blurred vision or other visual disturbances may affect the ability to drive or use machines. If blurred vision occurs after instillation, the patient must wait until the vision clears before driving or using machinery.

4.8 Undesirable effects

The following adverse reactions have been reported during clinical trials with MAXIDEX ophthalmic suspension and are classified according to the subsequent convention: very common ($\geq 1/10$), common ($\geq 1/100$ to $<1/10$), uncommon ($\geq 1/1,000$ to $<1/100$), rare ($\geq 1/10,000$ to $<1/1,000$), very rare ($<1/10,000$). Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

System Organ Classification	MedDRA Preferred Term
Nervous system disorders	<i>Uncommon</i> : dysgeusia
Eye disorders	<i>Common</i> : ocular discomfort <i>Uncommon</i> : keratitis, conjunctivitis, dry eye, vital dye staining cornea present, photophobia, vision blurred, eye pruritus, foreign body sensation in eyes, lacrimation increased, abnormal sensation in eye, eyelid margin crusting, eye irritation, ocular hyperaemia

Additional adverse reactions identified from post-marketing surveillance include the following. Frequencies cannot be estimated from the available data. Within each System Organ Class adverse reactions are presented in order of decreasing seriousness.

System Organ Classification	MedDRA Preferred Term (v. 18.1)
Immune system disorders	<i>Not known</i> : hypersensitivity
Endocrine disorders	<i>Not known</i> : Cushing's syndrome, adrenal insufficiency
Nervous system disorders	<i>Not known</i> : dizziness, headache
Eye disorders	<i>Not known</i> : glaucoma, ulcerative keratitis, intraocular pressure increased, visual acuity reduced, corneal erosion, eyelid ptosis, eye pain, mydriasis

4.9 Overdose

A topical overdose of MAXIDEX can be flushed from the eye(s) with lukewarm water.

Due to the characteristics of this preparation, no additional toxic effects are to be expected with an acute ocular overdose of this product or in the event of accidental ingestion of the contents of one bottle.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: corticosteroids. ATC code: S01BA01.

Mechanism of action

The exact mechanism of anti-inflammatory action of dexamethasone is unknown. It inhibits multiple inflammatory cytokines and produces multiple glucocorticoid and mineralocorticoid effects.

Pharmacodynamic effects

Dexamethasone is one of the most potent corticosteroids; with a relative anti-inflammatory potency greater than prednisolone or hydrocortisone.

Clinical efficacy and safety

The safety and efficacy of dexamethasone suspension/ointment have been established in adult clinical trials, published literature, and post-marketing surveillance.

Pediatric population

The safety and efficacy of dexamethasone suspension/ointment have not been studied in children; however, dexamethasone is reportedly safe for pediatric use, in general.

5.2 Pharmacokinetic properties

Absorption

After topical ocular administration, dexamethasone is detectable after 30 minutes in the aqueous humor and peaks at 90 to 120 minutes with a mean concentration of 31 ng/ml. Low but detectable concentrations are observed in the aqueous humor after 12 hours. Oral bioavailability of dexamethasone ranged from 70-80% in normal subjects and patients.

Distribution

After intravenous administration, the volume of distribution at steady state was 0.58 L/kg. In vitro, no change in human plasma protein binding was observed with dexamethasone concentrations from 0.04 to 4 µg/mL, with a mean plasma protein binding of 77.4%.

Biotransformation

After oral administration, two major metabolites were recovered which 60% of the dose was recovered as 6β-hydroxydexamethasone and up to 10% recovered as 6β-hydroxy-20-dihydrodexamethasone.

Elimination

After intravenous administration, the systemic clearance was 0.125 L/hr/kg. After i.v. bolus administration, 2.6% of the unchanged parent drug was recovered in the urine while up to 70% of the dose was recovered as identified metabolites. After systemic dosing, the half-life has been reported as 3-4 hours but was found to be slightly longer in males. This observed difference was not attributed to changes in systemic clearance but to differences in volume of distribution and body weight.

Linearity/non-linearity

Linear pharmacokinetics was observed after oral administration with doses between 0.5 to 1.5 mg where the AUC was less than proportional to the oral dose.

Pharmacokinetic/pharmacodynamic relationship(s)

A pharmacodynamic/pharmacokinetic relationship after topical ocular administration has not been established.

Special Population Pharmacokinetics

Pharmacokinetics of systemic dexamethasone did not significantly differ in renal-impaired patients when compared to normal subjects. Pediatric pharmacokinetics varied between age groups but wide interpatient variabilities were observed.

5.3 Preclinical safety data

Non-clinical data reveal no special hazard for humans, at the recommended clinical dose, based on conventional studies of repeated dose toxicity, genotoxicity, carcinogenic potential, or toxicity to reproduction and development.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Disodium phosphate, polysorbate 80, disodium edetate, sodium chloride, citric acid monohydrate and/or sodium hydroxide (to adjust pH), benzalkonium chloride, hydroxypropyl methylcellulose and purified water.

6.2 Incompatibilities

For information on the use of this product with contact lenses see Section 4.4 Special warning and precautions for use.

6.3 Storage Condition

Store below 30°C.

Store the bottle upright.

Keep the bottle tightly closed.

Do not use this medicine after the expiry date which is stated on the packaging.

Discard 4 weeks after first opening.

Keep this medicine out of the sight and reach of children.

6.4 Nature and contents of container

Sterile plastic bottle dispenser containing 5 ml.

7.0 Product Registration Holder

Novartis Corporation (Malaysia) Sdn. Bhd.

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No. 8, Jalan SS21/37, Damansara Uptown,

47400 Petaling Jaya, Selangor.

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